A Network Partitioning Algorithmic Approach for Macroscopic Fundamental Diagram Based Hierarchical Traffic Network Management

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***Abstract*—The existence of a Macroscopic Fundamental Diagram (MFD) in a network/subnetwork allows one to formulate hierarchical traffic management strategies. In order to achieve this, a robust and efficient network partitioning algorithm is needed. This research aims to create such an algorithm, where distinct MFD properties exist for each respective partition. The proposed four-step network partition approach utilizes the concept of lambda-connectedness and the technique of region growing, and, unlike prior studies, can work with partial traffic data. This research brings forth the following contributions: 1) an algorithmic approach that allows for incomplete traffic datasets as an input, and 2) an approach that does not require the user to arbitrarily pre-determine the number of necessary subnetworks. The proposed algorithmic approach can intuitively decide on the number of partitions based on the network connectivity and traffic congestion patterns.**

The proposed approach was implemented and tested on the regional planning network of Tucson/Pima County Arizona. The MFD related statistics for each subnetwork are presented and discussed. Numerical analysis on lambda choice and algorithm sensitivity regarding different data missing ratios were also performed and elaborated.

***Index Terms*—network partitioning, macroscopic fundamental diagram, hierarchical management strategy, lambda-connectedness, region growing.**

I. Introduction

Large-scale transportation networks need efficient traffic management and control schemes. Modeling a large transportation network is a complex task if traffic dynamics are studied and modelled at the link level. Moreover, centralized network control with such a detailed modeling approach would be computationally complex, which makes real-time implementation infeasible. A Macroscopic Fundamental Diagram (MFD) is a type of traffic flow fundamental diagram that relates the space-mean flow, the density, and the speed of an entire network. A MFD provides an approach to model urban traffic from a macroscopic point of view, which enables researchers and practitioners to model, monitor and control traffic in an aggregate manner without the need to work with detailed link information. MFD-based traffic control and management (Note that the terms "network controls and network management" are both used in this paper to reflect various network applications of the method.) aims to decrease inflows in areas with high densities of destinations and points and to manage accumulation in order to maintain the throughput flow in the city network at its maximum.

In a general, real-life transportation network, traffic patterns are typically heterogeneous as they may contain various types of roadway facilities, and the distribution of congestion is non-uniform and discontinuous with various spatial and temporal characteristics in the network. Recent findings show that the MFD shape of a heterogeneous network includes many scatter points when the transportation system becomes congested, which brings uncertainties and inaccuracies to the traffic state estimation [1-4]. Thus, partitioning a complete, heterogeneous urban transportation network into multiple homogeneous subnetworks is a prerequisite of MFD-based traffic flow control strategies, and is the objective that this research seeks to accomplish. This partitioning approach also contributes to the implementation of perimeter control, routing strategy, pricing, parking, and other management strategies, which all require the same homogenous condition.

In previous studies, various urban traffic network partitioning methods were proposed [5-7]. These approaches are mostly based on different graph cut methods and can only divide a network into two subnetworks of similar size at each iterative step. Some of these prior studies also reported high computational time [5]. The algorithm stops when a user-defined number of subnetworks is reached, although the method of determining the number of subnetworks has not been studied, and pre-defining the optimal number can be arbitrary and problematic. In addition, these studies assume there is complete traffic information over the entire network in order to calculate the correlation degree, or similarity, which requires accurate measurement of traffic conditions for all links. However, missing data is not a new problem for intelligent transportation systems. A clear illustration of this is the reality that, it is impossible to install loop detectors on every single road segment.

In a real-life traffic network, not all road segments have a detector installed, making the prior graph-cut based studies that assume perfect information impractical. This challenge of having partial data motivates the use of the lambda-connectedness concept combined with the dual graph construction step. Due to the definition of lambda-

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connectedness, any two vertices have a measurement of connectedness as long as there exists at least one path with data connecting them. This definition allows the connectedness to be defined and measured, even with partial data. By the definition of lambda-connectedness, any two vertices have a measurement of connectedness as long as there is at least one path with data connecting them. This is the fundamental difference when compared with the graph cut theory, in that it only deals with vertices and edges with or without connectivity. As in a real-life transportation network, only a few road segments may have detectors installed, such that for the other links, no correlation degree or similarity can be calculated or measured, and the application of the graph cut method based algorithm will be limited. Initially, the region growing approach was invented for image segmentation purposes [8, 9], but it can also be applied to address other problems related to uncertainty or incomplete information analysis. Based on the lambda-connectedness definition, the region-growing approach can be further designed and implemented in order to identify local homogeneous subnetworks.

In this paper, we propose a four-step algorithm used to partition a complete transportation network into multiple homogeneous subnetworks based on lambda-connectedness and region growing techniques. Firstly, an initial region growing model is built on the graph using lambda-connectedness as the edge weights, and local homogeneous subnetworks are extracted. Then, the region-growing technique is iteratively applied to the new graph which takes the subnetworks from the previous step as vertices. Considering that the definition of MFD exists on a network which requires a number of links and hierarchical control is always implemented at the subnetwork level where each subnetwork has a balanced size, a subnetwork merging algorithm is designed as the third step. Last but not the least, to make it practically applicable for traffic control and management purposes, a boundary building algorithm is proposed to find a clear convex boundary for each subnetwork.

The contributions of this research include but are not limited to:

- A 4-step network partition algorithm that extracts homogeneous subnetworks from the transportation network
- An algorithm that calculates and manages the size and number of subnetworks
- A lambda-connectedness technique that is capable of dealing with networks with incomplete information and calculating the vertices similarity from a global view
- A region growing technique, which uses heuristics and is a computationally efficient bottom-up approach, and will work effectively with real-life, large-scale transportation networks.

The remainder of this paper is organized as follows: section 2 reviews relevant prior research and background knowledge of lambda-connectedness and the region growing technique. Section 3 presents the four-step network partition algorithm. A case study built upon a large real-world network and numerical analysis on lambda determination and algorithm sensitivity regarding different data missing ratios are presented in section 4. Section 5 reviews and concludes this research and discusses potential future work.

II. Literature Review

A. Review of MFD and Network Partitioning

At the link level, fundamental diagrams (FDs) describe the relationship between flow, density, and speed [10]. Early studies concerning the extension of the FD concept to the network level with an optimum accumulation belonged to [11-13], and it wasn't until recently when the concept was theorized anew and named the Macroscopic Fundamental Diagram by Geroliminis and Daganzo [14]. This idea was furthered in a later paper, which showed that a theoretically reproducible relationship should arise if the traffic network met a few regularity conditions [15], and that such a relationship exists in Yokohama [3]. A MFD gives a macroscopic description of traffic by linking network flow and space mean densities, which can be used for modeling purposes, since detailed link information is usually very expensive and hard to measure [16, 17]. Based on the concept of MFD, elegant control strategies for subnetwork-based, hierarchical networks can also be designed to improve mobility and re-allocate congestion in homogenous regions [18-23]. Perimeter or gating control strategies, i.e. manipulating the transfer flows at the perimeter border of the urban region, utilizing the concept of the MFD have been introduced in [21-26]. Network level traffic state estimation utilizing MFD has been studied in [27]. Moreover, route guidance strategies with the utilization of MFD have been studied in [28-30]. In addition to online traffic control, studies have shown promising results concerning demand management with a MFD, like pricing policy in [31-33] and parking management in [34, 35].

Recent findings based on empirical or simulation data showed that heterogeneity has a strong impact on the shape of a MFD. Spatial distribution of link densities is a key factor that affects the shape of a MFD, and it also has an impact on the network-level traffic performance. The stability and observability of a well-defined MFD have been analyzed in many other simulation-based studies and experimental tests [1, 2, 4, 27, 36-43]. Networks with uneven distributions and inconsistent congestion may show traffic states that are well below the upper bound of a MFD and much too scattered to line on a MFD curve. In order to apply the MFD concept for modeling applications in heterogeneous transportation networks with multiple congestion centers, such networks should first be partitioned into a smaller number of homogeneous regions with low variance of link densities. This task is usually accomplished by graph partition techniques. Typically, graph partition problems fall under the category of NP-hard problems. Solutions to these problems are generally derived using heuristics and approximation algorithms and have two main categories, local and global. For a comprehensive review of research on graph partition problems, please refer to [44, 45].

Some urban transportation network partitioning methods were proposed in prior studies based on graph cut techniques.

Most of these previous studies seek to manage traffic signal control problems at a hierarchical level. Ma, et al. [7] tried to partition a traffic network into several sub-networks for control purposes using the spectral clustering method. The authors also developed a modularity metric to evaluate the partition schemes and determine the optimal result. Etemadnia, et al. [5] proposed a multi-way network partitioning methodology for distributed traffic management applications and presented two heuristics. For the Dallas-Fort Worth's regional network, the sparsest cut heuristic failed to converge at a reasonable running time. It showed that for a real-life transportation network, graph cut techniques are not the best solution for partitioning because of the computation complexity. These approaches mainly focus on the correlation of adjacent intersections, and do not take into account the effect of homogeneity in sub-networks, which makes them not suitable for MFD-based network partitioning.

Recently, network partitions used to support MFD-based network modeling became a hot research topic. Ji and Geroliminis [46] proposed a static network partition methodology with three consecutive algorithms, which are initial segmentation, merging, and boundary adjustment. Despite presenting an effective approach, it lays its foundation on the normalized cut algorithm, which can only divide a network into two equal sub-networks during each iterative step. Then based on the aforementioned research, Saeedmanesh and Geroliminis [47] further developed a partition methodology by proposing a new definition of link similarity based on running snakes. Later, they proposed a new 3-step optimization framework model and tested it on a large real-world network [48]. Pascale, et al. [49] proposed a spatiotemporal clustering method to detect homogeneous areas on both spatial and temporal dimensions. In each of these areas, a MFD can be defined.

Above all, these studies assume that there is full link information in order to calculate correlation degree or similarity and typically require accurate measurements of traffic conditions for all links. However, missing or low quality traffic data is a major problem for most transportation systems. A new way to perform network partitioning considering the practical partial data and information incompleteness should be developed for the purpose of implementing macroscopic traffic modeling, monitoring and control based on MFD.

B. Background Knowledge

1) Lambda-connectedness Definition

In graph theory, two vertices are said to be connected if there is a path between them. Lambda-connectedness [50], which is based on graph theory, is introduced to measure incomplete or fuzzy relations between two vertices. Graph theory only deals with vertices and edges with or without weights, and in order to define partial, incomplete, or fuzzy connectedness, one needs to assign a function to the vertex in the graph. Such a function is called a potential function, which can be used to represent the intensity of an image, the surface of a XY-domain, or the similarity of link density in a traffic network.

Assume a network system (G, ρ), where ρ is called a potential function of network G. A neighbor-connectivity will be first defined on a pair of adjacent points. Then, one can define the general connectedness between any two vertices. Assume $\alpha_\rho(x, y)$ is used to measure the neighbor-connectivity of x, y where x and y are adjacent vertices. In this research, the edge weight of adjacent vertices x, y, i.e. $\alpha_\rho(x, y)$ is measured by the difference of link densities. Thus, the potential function ρ is defined by the following equation:

$$\alpha_\rho(x, y) = 1/e^{(K_x - K_y)^2} \quad (1)$$

Where K_x and K_y are the link densities of vertices x, y.

In graph $G = (V, E)$ where V is the set of vertices and E is the set of edges, a finite sequence of $x_1, x_2, x_3, \cdots\cdots, x_n$ is called a path, if $(x_i, x_{i+1}) \in E$. The path-connectivity β of a path $\mathrm{P} = \mathrm{P}(x_1, x_n) = \{x_1, x_2, x_3, \cdots\cdots, x_n\}$ is defined as

$$\beta_\rho\big(\mathrm{P}(x_1, x_n)\big) = \min\{\alpha_\rho(x_i, x_{i+1}) | i = 1, \cdots\cdots, n-1\} \quad (2)$$

This equation implies that the connectivity is determined by the smallest connectedness between any adjacent pair of vertices. Taking a water tube as an example, the maximum volume that can go through the tube is determined by the thinnest part of the tube. Finally, the degree of connectedness (connectivity) of two vertices x, y with respect to ρ is defined as

$$C_\rho(x, y) = \max\{\beta_\rho\big(\mathrm{P}(x_1, x_n)\big) | \mathrm{P}\ is\ a\ simple\ path.\} \quad (3)$$

Through the equations above, the connectedness for a graph between any pair of vertices is determined by the highest path connectedness among all the paths connecting those two vertices. By this definition, any two vertices will have a connectedness value as long as there is at least one path connecting them. Therefore, lambda-connectedness is a method to define a partial, incomplete, or fuzzy connectedness.

For a given $\lambda \in [0, 1]$, points x, y are said to be $\lambda - connected$ if,

$$C_\rho(x, y) \geq \lambda. \quad (4)$$

2) Region Growing Technique

Region growing is a simple, region-based image segmentation method. Region-based methods rely on the postulate that neighboring pixels within one region have similar values. This approach to segmentation examines neighboring pixels of initial seed points and determines whether or not the pixel neighbors should be added to the region. The general procedure is to compare one pixel to its neighbor(s). If a criterion of homogeneity is satisfied, the pixel is said to belong to the same class as one or more of its neighbors. With this mechanism, one can simply define the threshold of homogeneity measurements and the algorithm will partition the network according to this criterion, without the need of pre-defining the number of final clusters.

By combining the definition of lambda-connectedness and region growing techniques, we developed a 4-step network partition algorithm to divide a network with only partial data into multiple sub-networks based on the minimum degree of connectedness allowed. The following section will present the methodology in detail.

III. METHODOLOGY

Our main objective is to partition a heterogeneous complete transportation network into multiple homogeneous subnetworks so that a well-defined MFD can be observed in each subnetwork. More specifically, we seek to develop a partitioning algorithm that can complete the following:

- Identify locally homogeneous components with low variance in link densities in each subnetwork to ensure a well-defined MFD
- Extract subnetworks with large inter-subnetwork density variance to ensure distinct inter-subnetwork congestion patterns
- Limit the number of subnetworks with balanced network sizes
- Build smooth boundaries for subnetworks to facilitate traffic management strategies

For data preparation, transportation networks are represented by a graph where the vertices are the links and the edges are given by the network topology. Applying the region-growing algorithm, the transportation network partition is equivalent to a decimation of the graph.

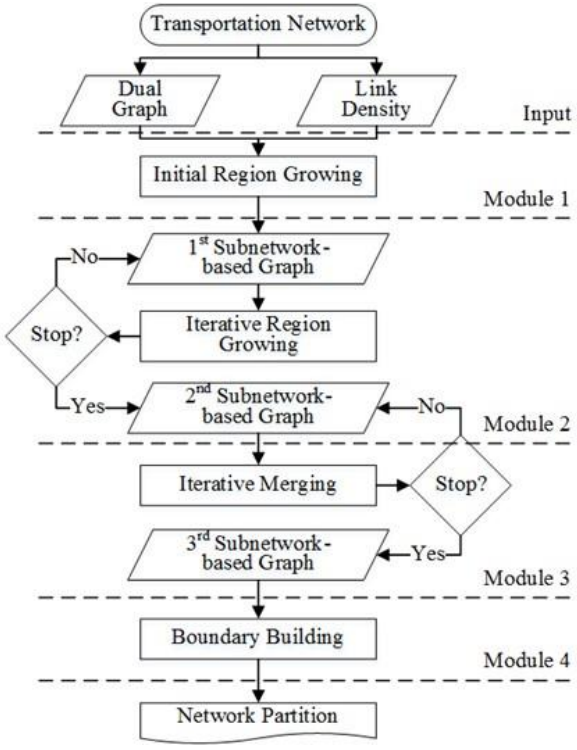


Figure 1 Workflow Diagram

The proposed overall workflow is shown in Figure 1which mainly includes the four modules listed below.

- **Module 1 - Initial Region Growing Module**, which is applied to identify the link-level locally homogeneous components, to accomplish objective 1
- **Module 2 - Iterative Region Growing Module**, which iteratively converts subnetworks from step 1 into vertices, reconstructs the edge weights, builds a subnetwork-based graph, and performs region growing on the subnetwork-based graph
- **Module 3 - Iterative Merging Module**, which is applied to iteratively merge the smallest subnetwork to its most connected subnetworks, in order to meet objective 3
- **Module 4 - Boundary Adjustment Module**, which is used to build the boundaries for these subnetworks, in order to meet objective 4

The following sections will describe each module in details.

A. Initial Region Growing

In the first module, we apply the region-growing algorithm to extract all local homogeneous subnetworks from the global network to meet the requirements of the first and second research goals.

The first step in the region growing algorithm is to select a seed vertex to start growing the subnetwork. Lambda-connectedness is reflexive, symmetric, and transitive, making it is an equivalence relation [9]. This indicates that the network partition result has no relationship with the seed selection method. In this research, the vertex with the largest connectedness is selected as the seed. Afterwards, a Depth First Search (DFS) algorithm is applied to realize the initial region growing. The value of lambda determines if the search on a branch shall be stopped or continued. Details of the DFS algorithm, together with the pseudocode, are shown below.

Algorithm: Initial Region Growing

I. *Network Initialization*
$i = 0$: cluster IDs
$G = \{v|\text{vertices in } Graph\}$: set of vertices
$\Lambda(v) = \{g|\text{vertices that are spatially adjacent to } v\}$
$\iota(v) = 0$: label each vertex as zero
$\omega(\theta, \mathrm{v})$: lambda-connectedness between θ and v
$\Omega(g) = sum(\omega(g, v)|v \text{ in } \Lambda(g))$

II. *Loop through all vertices*
for $g \in G$:
  while $G! = []$
    $\theta = \{g|\Omega(g) = \max(\Omega(G))\}$
    if $\Omega(\theta) < \lambda$ **do**
      pass
    else do
      for v *in* $\Lambda(\theta)$
        if $\min(\iota(\theta), \omega(\theta, \mathrm{v})) > \omega(\mathrm{v})$
          $\omega(\mathrm{v}) = \min(\omega(\theta), \omega(\theta, \mathrm{v}))$
        $G = G - v$
        $V = V + v$
        $\theta = \{g|\max(\omega(g, v), \mathrm{v} \text{ in } \mathrm{V})$
    $\phi(i) = V$
    $i = i + 1$

The DFS algorithm searches every vertex only once, which means that the designed algorithm has a complexity of only $o(n)$ and is more efficient than most existing methods in this regard.

B. Iterative Region Growing

After the first module, we now have multiple locally homogeneous subnetworks. However, although it lacks link-level homogeneity, a subnetwork usually has some adjacent subnetworks with similar average densities due to congestion spatial propagation, which means that they are homogeneous at

a higher data aggregation level. An iterative region-growing module is designed to capture the congestion propagation phenomena and further grow the regions at the subnetwork level.

These locally homogeneous subnetworks can be connected by iteratively growing these subnetworks with the same potential function and search algorithm in section A. In the second module, we iteratively apply the region-growing technique on the subnetwork from section A. Each subnetwork from the previous iteration is converted to a vertex, and if there is at least one edge connecting any two subnetworks in the previous network, then the two vertices representing them are connected by one edge. Take Figure 2 for example. Black dots represent links in the transportation network, and after the Initial Region Growing, the transportation network is partitioned to three subnetworks, A, B , and C . At the subnetwork-based graph, three vertices represent them, and since they are adjacent with each other, three edges are created to connect them. The connectedness is calculated using the same equation, and the density here is found by taking the subnetwork average density.

$$K_s = \frac{\sum_{i=1}^{n_s} l_i lane_i K_i}{\sum_{i=1}^{n_s} l_i lane_i} \quad 5$$

Where K_s is the average density of subnetwork s and $l_i, lane_i, K_i$ are the link length, lane number and density respectively of link i in subnetwork s. K_s is equivalent to the total vehicles in the subnetwork s divided by lane length (sum of link length weighted by lane number).

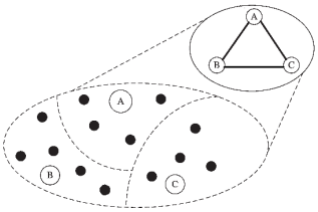


Figure 2 Subnetwork to Graph Conversion Method

In a practical situation, the lambda-connectedness approach could run into an exception when a sub-graph is surrounded by missing data links. As a result, the similarity between certain links may become zero. This issue is addressed in Module 2. The isolated sub-graph is converted to a vertex in the hierarchical graph, and similarity values are calculated for the edges connecting the isolated sub-graph and other parts.

C. Iterative Subnetwork Merging

After Iterative Region Growing, we obtain plenty of subnetworks which are locally homogeneous and spatially compact. Based on previous researches from simulation and empirical data, homogeneous regions with proper size usually show a good MFD curve. In research [46], each subnetwork consists of about 100 links and they show a good representation of the MFD curve. However, the method to calibrate this optimal number is not performed and is beyond the scope of this research. In this module, 100 is taken as the minimum number of links for a subnetwork.

In this module, an iterative merging algorithm is applied to form new subnetworks based on the initial results in order to achieve the third research goal. During each iteration, a subnetwork based graph is built and edge weights are calculated using the same method in section B. To ensure homogeneity in subnetworks, the smallest subnetworks are merged to their adjacent subnetwork with the largest connectedness, since having the largest connectedness indicates the smallest gap of densities. This iteration will keep running until the size of the smallest subnetwork has a number of links larger than a predefined value, which is 100 for this study.

This iteration method also provides an intuitive treatment for the missing data. Each of the missing data links is converted to a vertex that has a density equal to zero in the first three modules, but these vertices are not processed until the last module. Each of these vertices contains only one missing data link and is treated as the smallest cluster. The proposed algorithm would merge such a vertex to its adjacent subnetwork with the largest connectedness due to similar density values.

D. Boundary Building

After these three steps, we get a rough partitioning result. However, these subnetworks are non-smooth because some subnetworks may contain links that overlap with each other. To overcome this issue, we developed a straightforward GIS-based boundary building method to make the subnetwork boundaries clearer and easier for traffic management purposes.

First, a convex boundary is built for every subnetwork. Then, the relations of links to subnetworks are rebuilt. If a link is inside the boundary of a certain subnetwork, this link belongs to the subnetwork. However, some subnetworks may share some links after the convex boundaries are built. For each overlapping area, there will be two intersection points. A straight line is drawn between the two intersections and the shared area will be separated into two areas. Each of the links will be assigned to one of the overlapped subnetworks. By doing so, the new boundaries are adjusted to maintain the convexity of the boundary for both subnetworks.

IV. Case Study

In this section, the proposed algorithm is implemented on a large real-world transportation network. The real-life large scale network partitioning results and subnetwork MFD's are proposed to demonstrate its effectiveness. Furthermore, sensitive analysis is performed to determine the best lambda value to be utilized. Finally, the robustness of this algorithm is tested with various missing data ratios.

A. Data Preparation

For this research, we used the Tucson transportation network as a test bed. This network has over 38,000 undirected links. A 24-hour time-dependent traffic demand was applied to the network on the DynusT simulation platform, which reproduced the spatial and temporal distribution of congestion. The simulation model has been calibrated with traffic counts and time-dependent speed profiles as calibration targets [51] and further examined by the local transportation agency.

The transportation network dual graph is built as an

undirected graph (G, ρ). Each vertex i in G represents an undirected link in the network and has a density value K_i of the link at a certain time of day. The adjacency list measures the neighboring relation between each pair of links. The edge weights are calculated by Equation 1.

B. Partition Results and MFD's Analysis

The effectiveness of the proposed algorithm is tested on the network with traffic information at time t = 8:00 AM, when the network is highly congested and shows different levels of congestions in different sub-areas. In this research, we used three criteria to evaluate the partition result. Each criterion is related to one objective in section III.

- **Variance of Density**: A measurement of homogeneity within each subnetwork. This criterion refers to the first research goal.
- **NS**: A measurement metric [46] that measures the density difference between a subnetwork and its most similar adjacent subnetwork. Here we follow the same definition for comparison purposes. A small NS value implies a well partitioned subnetwork. This criterion is used to evaluate the algorithm efficiency for the second research goal.
- **Links Count**: The number of undirected links in a subnetwork that is associated with the subnetwork size. This is an evaluation criterion relating to the third research goal.

The four-module network partition algorithm finished within 10 minutes on the test network. The algorithm was also tested on the Houston TX regional network, and it gave satisfactory results within 15 minutes. From these two tests of the algorithm's computational efficiency, the algorithm proposed in this paper is fast enough for real-time network control applications.

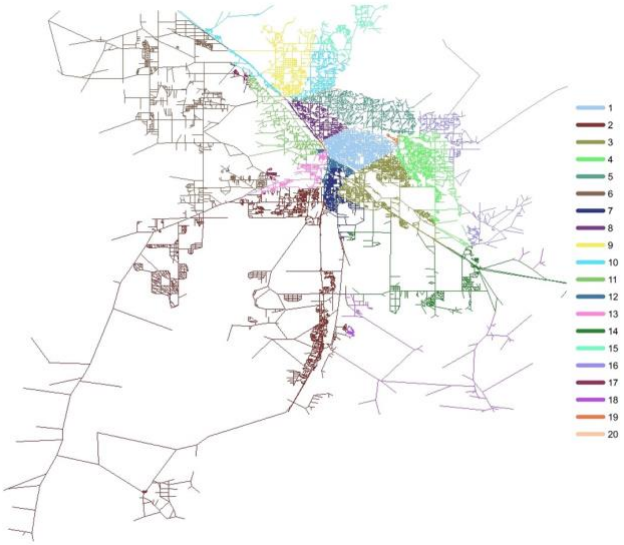


Figure 3 Network Partition Result with Lambda = 0.1

The network partitioning result on the Tucson AZ network can be found in Figure 3. In the end, the algorithm generates 20 subnetworks, each of which is found to have a clear boundary and ensures the applicability of MFD-based traffic management and control strategies. The first three modules are developed to meet different objectives of network partition problem in order to ensure a well-defined MFD curve. The fourth module is not constrained by homogeneity requirement. However, from a comparison of aforementioned criterion on the results of Module 3 and Module 4, the subnetworks do not change much.

Table 1 presents the partitioning results corresponding to the three criteria defined above. The link counts range from 326 to 5206, which as discussed in section III.C, is in a reasonable range to ensure the existence of MFDs. The average link density varies from 0.8 to 26.22veh/ln/mile with small variances indicating the inner-subnetwork homogeneity. The average NS value is 0.5890, which indicates a well partitioned result when compared with the NS values in [46] (NS is improved from 0.7442, 0.6865 to 0.6210 in three steps).

Table 1 Statistics of Subnetworks with Lambda = 0.1

ID	**1**	**2**	**3**	**4**	**5**
#Link	5206	4382	3444	3028	2042
Variance	5.06	2.66	10.6	3.76	6.53
NS	0.8	0.68	0.92	0.85	0.95
ID	**6**	**7**	**8**	**9**	**10**
#Link	1774	1404	1355	1281	1232
Variance	3.51	5.28	1.74	4.12	5.31
NS	0.51	0.78	0.79	0.82	0.12
ID	**11**	**12**	**13**	**14**	**15**
#Link	1228	1172	1170	1133	1004
Variance	76.5	18.9	3.8	3.22	10.5
NS	0.28	0.25	0.53	0.64	0.89
ID	**16**	**17**	**18**	**19**	**20**
#Link	726	713	689	671	326
Variance	12.9	11.1	7.85	1.11	12.6
NS	0.43	0.37	0.47	0.56	0.14

We also investigated the existence of MFDs for all subnetworks, and the results are summarized in Figure 4 below. They look similar, so only those which exhibited distinct MFD characteristics are shown in Figure 4 . With the network partitioning result, flow and density data were aggregated at 5 min intervals based on Edie's method [52]. For each subnetwork, the pattern of a MFD is rather clear and intuitive, which indicates a satisfactory partitioning result. The MFDs of Regions 7 and 8 have different maximum output flows from those of Regions 18 and 19. Regions 7 and 8 are located around the downtown area, where there are lots of local streets, and the maximum output flow is around 200-400 veh/ln/hr. Regions 18 and 19 are in suburban areas and contain higher proportions of high class roads, so the maximum output flow, which are 600 and 800 veh/ln/hr respectively, are much higher than those of Regions 7 and 8.

The critical density of these MFDs is about 20-30 veh/ln/mile, which is slightly smaller than the result from

Geroliminis and Daganzo [3] (about 35veh/ln/mile) and Buisson and Ladier [1] (about 55 veh/ln/mile), but similar to that of Mahmassani, et al. [43] (about 25 veh/ln/mile). From a data source perspective, considering that the results of Geroliminis and Daganzo [3] and Buisson and Ladier [1] are from loop detector data, while this research and Mahmassani, et al. [43] used simulation data to plot MFDs, the results seem to be comparable and explainable, as usually loop detectors are only located on major roads, while simulation-based research uses all link data including minor streets, and its density is averaged at a network level and will be lower than the former approach. This finding agrees with the conclusions in [53].

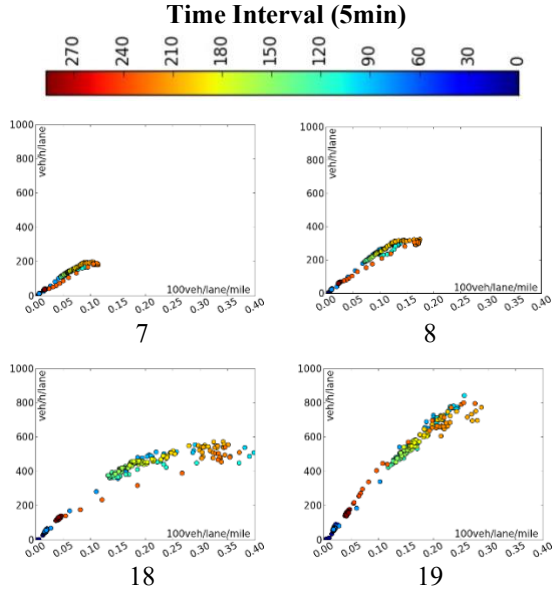


Figure 4 Subnetwork MFD's

The macroscopic flow and density relationship from the time dimension was studied. The gradient color in the MFD curves, from 1 to 288, represented the time dimension, which meant time from 0:00 to 23:55. Larger subnetworks showed well-defined MFDs, while there were some scatters in smaller subnetworks as shown in Figure 4. Some subnetworks showed capacity drop, i.e. scatters, during the AM peak. This property can be utilized to identify the subnetworks that should be protected during peak hours. In this case, Regions 4, 11 and 16 would be protected. MFD-based control and management strategy, like routing, gating control, etc., can be implemented to constrain the accumulation of protected subnetworks and decrease inflows in these subnetworks in the decreased part of an MFD to maintain the throughput flow in the city at its maximum.

Some of these MFDs look very similar, but it is improper to group them together because of two reasons, 1) they are not spatially adjacent and 2) they have different levels of congestion during the same time window. However, the shapes and critical values of the MFDs will not change much if adjacent uncongested clusters are grouped. This interesting finding is agreed with that of Cassidy, et al. [54]. The congestion distribution will change over time, further improvement of the proposed approach to deal with dynamic situation is of future concern.

C. *Lambda Choice Considerations*

As discussed, lambda is a key parameter in the proposed algorithm and its value directly determines the network partitioning result. This section performs sensitivity analysis over different values of lambda, with the hope of presenting a procedure for the determination of an optimal value to be used. Corresponding to the evaluation criteria in section B and referring to sensitivity analysis in the image processing area [50], two criteria are proposed to determine the proper lambda value.

- **Max Subnetwork Size**: In general, we would like to have similar subnetwork sizes in the partitioned result
- **Subnetwork Density Variance**: This criterion measures the difference of the average densities in different subnetworks. Larger Subnetwork Density Variance indicates a larger difference of congestion between subnetworks, and is desired.

Figure 5 presents the lambda sensitivity analysis results, with lambda values 0.1, 0.01, 0.001, 0.0001, 0.00001 tested. They correspond to allowed link density differences of 1.52, 2.15, 2.63, 3.03 and 3.39 vehicle/lane/mile, respectively. The sensitivity test results are shown below:

- Based on the first criteria on max subnetwork size, lambda value 0.1 is able to achieve the best partitioning result, and while lambda decreases, the results become less satisfactory.
- Based on the second criteria on Subnetwork Density Variance, the best result is achieved when lambda = 0.0001 where the Subnetwork Density Variance is the highest among all tested lambda. The second-best result is when lambda takes value of 0.1.
- Combining 1 and 2, we take 0.1 as the optimal value to be used in our research, as it leads to the best partitioning result with regard to the max subnetwork size, and the second-best result with regard to the Subnetwork density variance. However, researchers can choose the optimal lambda value based on their own objectives and criteria.

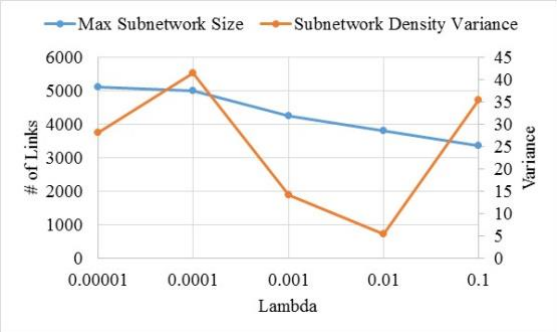


Figure 5 Sensitivity Analysis of Lambda

D. *Partial Data Analysis*

As explained in section II.B.1), one unique advantage of lambda-connectedness is that it can be used to measure the connectivity of any two vertices in a network even if there is only partial data available. We proposed a criterion to evaluate the similarity of two partition results with and without data

missing. We call the network partition result using full link information ***Full Result,*** and the network partition result using partial data ***Partial Result***. For each subnetwork in Full Result, we find a subnetwork in Partial Result that shares the most links and calculate the percentage of links that they share, and use the ratio of shared links to measure the subnetwork similarity between the Full Result and Partial Result.

$$\text{Similarity}_i = \left(\frac{n_i}{\max(N_i, N_i^p)}\right)^2 \quad 6$$

Where, n_i is the number of shared links in subnetwork i, and N_i, and N_i^p are the total number of links in subnetwork i from Full Result and Partial Result respectively. To measure the total similarity between Full Result and Partial Result, we further use the metric below.

$$\Delta = \frac{\sum_i^n N_i \times \text{Similarity}_i}{N} \quad 7$$

Where n is the number of subnetworks and N is the total number of links in the whole network.

Figure 6 shows the similarity with different data missing ratios, and we can observe that as the missing rate increases, the similarity between Full Result and Partial Result becomes smaller, indicating that the network partitioning result becomes less satisfactory, which is rather logical. If the acceptable similarity is set to be 50%, meaning the overlap between a new partitioning result and original solution needs to be higher than 50% to make it acceptable, the maximum data missing rate allowed can be observed to be 60%, indicating that the proposed algorithm needs to have at least 40% data coverage in order to achieve a good partitioning result. Considering data availability on most freeway or major arterials from various data collection mechanisms, combined with a properly designed data imputation algorithm, this data requirement should be easily satisfied, which ensures the practical applicability and effectiveness of this proposed algorithm.

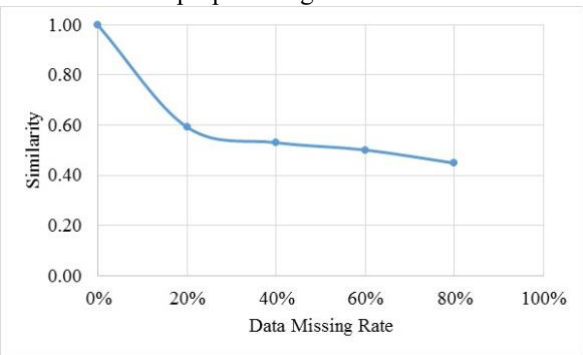


Figure 6 Similarity with Different Data Missing Rate

V. Discussions

This paper aims to address a network partition problem in large-scale hierarchical traffic control and management. A network-partitioning algorithm based on lambda-connectedness and region growing techniques is proposed to capture the spatial congestion heterogeneity as a prerequisite of applying a MFD for network-wide traffic control and management purposes. Compared with the methods proposed in prior studies, the unique advantages of the proposed four-step algorithm lie in its capability to work with partial data, which is very common in reality, and does not require a pre-defined number of subnetworks as a modeling input, which could be arbitrary and problematic. The numerical analysis performed in this research, including statistics analysis, MFDs for each subnetwork, sensitivity analysis on lambda determination, and different data missing ratios all suggest satisfactory results. MFD's in each subnetwork showed well-defined concave curves, which shed light on its application in hierarchical MFD-based traffic control and management. Heterogeneous transportation network can be partitioned into homogeneous subnetworks to ensure the existence of MFDs, and dynamics of subnetworks can be modeled and further controlled in an aggregate manner.

It's noteworthy that in the initial region growing module and iterative region growing module, we found that link type affects the partition result, and links of the same type are more likely to be grouped into the same region. In addition, Xie, et al. [55] also found network heterogeneity, road capacity, and the attraction strength of a subnetwork all have impacts on the shape of a MFD. Network partitions considering heterogeneity in both traffic congestion and network topology should be of future concern. Furthermore, in the iterative merging module, 100-link was used as the empirical threshold for the minimal subnetwork size. Future research is needed to calibrate such a threshold by considering the network homogeneity and implementation considerations with regard to hierarchical traffic network control.

This research is a bottom-up approach to forming network partitions, while other existing approaches are top-down approaches. Hierarchical approaches, by applying region growing techniques to extract locally homogeneous subnetworks and to further implement graph cut techniques to balance the size of subnetworks will be the future research direction. In addition, we envision that as traffic evolves by time of the day, time-varying optimal lambda and NS values may exist, warranting the exploration and development of a dynamic network partition algorithm.

VI. Acknowledgement

The authors would like to thank Dr. Robert Tung from RST International Inc. for his help with the DynusT simulation model and Dr. Steven Feng from UA for background knowledge about image processing. Transportation network and demand data from Pima County Association of Governments (PAG) are appreciated.

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